

NICHOLAS MUCHELEMBWA

Backend & API Software Engineer | Cloud & AWS | Python / FastAPI / Java

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PROFESSIONAL SUMMARY

Backend and API software engineer with 5+ years building secure, scalable RESTful APIs and cloud architectures for healthcare, finance, and resource-sector organizations. Specializes in Python (FastAPI), Java, and AWS, delivering services that power telemedicine platforms, enterprise reporting tools, and data-driven applications. Experienced leading project delivery across the full software lifecycle, coordinating cross-functional teams, managing scope and timelines, and aligning architecture and product stakeholders. AWS Certified Solutions Architect, Certified Kubernetes Administrator, and PMP-certified, with a record of reducing manual processes, improving system performance, and shortening deployment cycles.

TECHNICAL SKILLS

Languages: Python, Java, C#, PHP, JavaScript, SQL

Backend & APIs: FastAPI, REST API design, microservices, Laravel Lumen, .NET / Blazor, secure backend services

Cloud & DevOps: AWS (S3, Athena, SageMaker, Kendra, Translate, Lambda, API Gateway, EKS), Docker, Kubernetes, CI/CD

Data & ML: Machine Learning, TensorFlow, Power BI, data processing, workflow automation

Practices: Software Development Lifecycle (SDLC), Agile, system integration, third-party API integration, production support

PROFESSIONAL EXPERIENCE

Orbis International

Lusaka, Zambia

API Software Engineer

July 2024 - Present

- Led the design and development of scalable RESTful APIs supporting a global telemedicine and AI-enabled healthcare platform.
- Improved API reliability and performance through cloud-native practices and consistent interface contracts.
- Built and maintained backend services using Python (FastAPI) and Java across multiple deployment environments.
- Integrated third-party services with external vendors and partners, improving system interoperability.
- Contributed to backend architecture decisions and CI/CD workflows alongside architects and cross-functional teams.

Software Developer (Backend & Cloud)

February 2023 - June 2024

- Developed and maintained backend services and RESTful APIs using Python, Java, and FastAPI for cloud-based business applications.
- Implemented data processing and automation solutions with AWS services (S3, Athena, Translate, Kendra, SageMaker), improving data accessibility and operational efficiency.
- Designed and delivered Power BI dashboards supporting analytics, reporting, and data-driven decision-making.
- Built backend integrations for internal tools and external-facing systems, improving reliability and interoperability.
- Contributed across the full software development lifecycle: requirements analysis, development, testing, deployment, and production support.

First Quantum Minerals

Solwezi, Zambia

Software Developer

April 2021 - March 2023

- Developed and maintained internal business applications using C# (.NET / Blazor) to support mining operations and reporting workflows.
- Designed and delivered Power BI reports and dashboards, improving visibility into operational and production data.
- Built Python-based automation scripts that reduced manual processes across data and reporting tasks.
- Applied machine learning techniques to improve prediction and analysis of mining BCM (Bank Cubic Meter) data.

National Institute of Public Administration (NIPA)

Lusaka, Zambia

Software Developer

October 2020 - April 2021

- Implemented and enhanced application features to improve system functionality, usability, and performance.
- Introduced Python automation solutions that reduced repetitive tasks and improved operational efficiency.
- Worked across a mixed stack (PHP, JavaScript, Java, JSP, Python) to deliver reliable, maintainable software.

The Global Air Drone Academy

Software Engineer

July 2020 - April 2021

- Developed Python scripts to automate drone operations and support flight-related workflows.
- Assisted in building and testing drone systems, gaining hands-on hardware and software integration experience.
- Trained students on safe drone operation and compliance with aviation safety regulations.

Zambia Industrial Commercial Bank (ZICB)

Lusaka, Zambia

Information Technology Intern

August 2020 - October 2020

- Developed a user request management system using C# and PHP to support internal service workflows.
- Assisted in bringing Wincor ATM systems into production and provided support for Flexcube card-link systems within a regulated banking environment.

CERTIFICATIONS

- AWS Certified Solutions Architect – Associate
- AWS Certified Machine Learning – Specialty
- Certified Kubernetes Administrator (CKA)
- TensorFlow Developer Certificate
- Project Management Professional (PMP)®
- Responsive Web Design (freeCodeCamp)

EDUCATION

MSc, Data Science

ZCAS University, Lusaka, Zambia

2026 - 2027 (In Progress)

BSc, Computer Science (Summa Cum Laude)

Mulungushi University, Kabwe, Zambia

2016 - 2020

HONORS & AWARDS

- Zanaco University Challenge – Winner